

Positive Functional Equations for Cylindric Partition Generating Functions: the R -Relation Propagation Theorem

(Seed 6, Layer 2, Round 2)

Loop Experiment

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Abstract

For rank-3 cylindric partitions we show that every profile containing a zero part admits an explicit *manifestly positive* functional equation (an “ R -relation”) derived from the Corteel–Welsh system, for every level d . Consequently, bivariate coefficient positivity of $G_c(z, q) = (zq; q)_\infty F_c(z, q)$ for *all* profiles reduces to the finite set of *all-positive* profiles ($c_0, c_1, c_2 \geq 1$). At $d = 5$ this core is exactly the two profiles Warnaar treated by computer algebra, explaining the structure of the $d = 5$ proof and showing the propagation mechanism is not d -specific. At the bounded-polynomial level we obtain the exact identity $Q_n^c = q^n Q_n^{\text{head}} + (1 - q^{\ell n}) \sum_i q^{(i+1)n-1} Q_{n-1}^{b_i}$ and formulate a cross-profile injection inequality (INJ) that, together with core positivity, implies the full positivity conjecture; INJ is verified computationally for $d \in \{4, 5, 6, 7, 8, 10, 11\}$. As a corollary of the R -relation calculus, the $d = 2$ case is solved exactly: $Q_n^{(1,1,0)} = q^{n^2}$ and $Q_n^{(2,0,0)} = q^{n(n+1)}$.

1 Setup

Fix $r = 3$. For a profile $c = (c_0, c_1, c_2)$ with $d = c_0 + c_1 + c_2$ let $F_c(z, q)$ be the cylindric partition generating function and

$$G_c(z, q) := (zq; q)_\infty F_c(z, q),$$

which is invariant under cyclic rotation of c . Writing $I_c = \{i : c_i > 0\}$ and $c(J)$ for the Corteel–Welsh shifted profile ($c_i(J) = c_i - 1$ if $i \in J$, $i - 1 \notin J$; $c_i + 1$ if $i \notin J$, $i - 1 \in J$; else c_i ; indices mod 3), the Corteel–Welsh functional equation reads, in G -form,

$$G_c(z) = \sum_{\emptyset \neq J \subseteq I_c} (-1)^{|J|-1} (zq; q)_{|J|-1} G_{c(J)}(zq^{|J|}). \quad (1)$$

For $|I_c| = 1$, say $I_c = \{j\}$, this is simply $G_c(z) = G_{c(\{j\})}(zq)$, with no signs. For $|I_c| \geq 2$ the signs alternate; this is the obstruction to naive positivity propagation.

For $\ell = \gcd(d, 3)$ the bounded polynomials are $Q_{n,c}(q) = (q^\ell; q^\ell)_n [z^n] G_c(z, q)$, and the conjecture (CDU 2020 / Warnaar 2023) asserts $Q_{n,c}(q) \geq 0$ coefficientwise for $d \not\equiv 0 \pmod{3}$.

2 R -relations and the Substitution Lemma

Definition 1. A profile c admits an R -relation if there exist a profile $\text{head}(c)$ and a finite list $b_1, \dots, b_L$ of profiles (the *tail*) such that

$$G_c(z) = G_{\text{head}(c)}(zq) + \sum_{i=1}^L zq^i G_{b_i}(zq^{i+1}). \quad (2)$$

Every coefficient on the right of (2) is a nonnegative combination of coefficients of G 's at other profiles: the relation is manifestly positive.

Lemma 2 (Substitution Lemma). *Let $|I_c| = 2$ with $I_c = \{j_1, j_2\}$, so (1) reads*

$$G_c(z) = G_{c(\{j_1\})}(zq) + G_{c(\{j_2\})}(zq) - (1 - zq) G_{c(\{j_1, j_2\})}(zq^2).$$

Suppose $c(\{j_2\})$ admits an R -relation whose head equals $c(\{j_1, j_2\})$ up to rotation, with tail $b_1, \dots, b_L$. Then c admits the R -relation

$$G_c(z) = G_{c(\{j_1\})}(zq) + zq G_{c(\{j_1, j_2\})}(zq^2) + \sum_{i=1}^L zq^{i+1} G_{b_i}(zq^{i+2}),$$

i.e. $\text{head}(c) = c(\{j_1\})$ and $\text{tail}(c) = [c(\{j_1, j_2\})] ++ \text{tail}(c(\{j_2\}))$.

Proof. Substitute $w = zq$ into the R -relation of $c(\{j_2\})$: $G_{c(\{j_2\})}(zq) = G_{c(\{j_1, j_2\})}(zq^2) + \sum_i zq^{i+1} G_{b_i}(zq^{i+2})$, using rotation invariance of G to identify the head with $c(\{j_1, j_2\})$. Inserting this into the displayed CW equation, the term $+G_{c(\{j_1, j_2\})}(zq^2)$ combines with $-(1 - zq)G_{c(\{j_1, j_2\})}(zq^2)$ to give $+zq G_{c(\{j_1, j_2\})}(zq^2)$. No other cancellation occurs. $\square$

3 The two-family construction

We now show the hypothesis of Lemma 2 can always be arranged for zero-containing profiles. By rotation invariance it suffices to treat $c = (a, 0, b)$ and $c = (a, b, 0)$ with $a, b \geq 1$, plus the base case $(d, 0, 0)$.

Theorem 3 (R -relations for all zero-containing profiles). *Let $d \geq 1$.*

(0) $c = (d, 0, 0)$: $I_c = \{0\}$, and (1) is already the R -relation $G_{(d, 0, 0)}(z) = G_{(d-1, 1, 0)}(zq)$ with empty tail.

(A) $c = (a, 0, b)$, $a, b \geq 1$: c admits the R -relation with

$$\text{head} = (a-1, 1, b), \quad \text{tail} = [(a, 1, b-1), (a+1, 1, b-2), \dots, (a+b-1, 1, 0)].$$

(B) $c = (a, b, 0)$, $a, b \geq 1$: c admits the R -relation with

$$\text{head} = (a, b-1, 1), \quad \text{tail} = [(a-1, b, 1)] ++ \text{tail}((b+1, 0, a-1)),$$

where for $a = 1$ the profile $(b+1, 0, 0)$ has empty tail, so $\text{tail}((1, b, 0)) = [(0, b, 1)]$.

Proof. (A) Induction on b . Here $I_c = \{0, 2\}$, $c(\{0\}) = (a-1, 1, b)$, $c(\{2\}) = (a+1, 0, b-1)$, $c(\{0, 2\}) = (a, 1, b-1)$. Take $j_1 = 0$, $j_2 = 2$ in Lemma 2. If $b = 1$, then $c(\{2\}) = (a+1, 0, 0)$ and by (0) its head is $(a, 1, 0) = c(\{0, 2\})$ with empty tail; the lemma gives head $(a-1, 1, 1)$, tail $[(a, 1, 0)]$. If $b \geq 2$, by induction $c(\{2\}) = (a+1, 0, b-1)$ has head $(a, 1, b-1) = c(\{0, 2\})$ exactly, and tail $[(a+1, 1, b-2), \dots, (a+b-1, 1, 0)]$; the lemma prepends $(a, 1, b-1)$, giving the stated tail.

(B) Here $I_c = \{0, 1\}$, $c(\{1\}) = (a, b-1, 1)$, $c(\{0\}) = (a-1, b+1, 0)$, and $c(\{0, 1\}) = (a-1, b, 1)$. Take $j_1 = 1$, $j_2 = 0$. If $a \geq 2$, rotate: $c(\{0\}) = (a-1, b+1, 0) \sim (b+1, 0, a-1)$, which by (A) has head $(b, 1, a-1) \sim (a-1, b, 1) = c(\{0, 1\})$ up to rotation, and tail $\text{tail}((b+1, 0, a-1))$; the lemma gives head $c(\{1\}) = (a, b-1, 1)$ and tail $[(a-1, b, 1)] \uparrow \text{tail}((b+1, 0, a-1))$. If $a = 1$, $c(\{0\}) = (0, b+1, 0) \sim (b+1, 0, 0)$ with head $(b, 1, 0) \sim (0, b, 1) = c(\{0, 1\})$ by (0), empty tail; the lemma gives head $(1, b-1, 1)$, tail $[(0, b, 1)]$. $\square$

Remark 4. R -relations are not unique: e.g. for $c = (a, 1, 0)$ the choice $j_2 = 1$ yields a second valid relation with head $(a-1, 2, 0)$ and a shorter tail. Both variants were verified numerically; non-uniqueness is exploited in Section 6.

Remark 5 (Consistency with the literature). At $d = 5$, Theorem 3 reproduces the three positive functional equations of Corteel–Dousse–Uncu (their eqs. 3.17–3.19) *exactly*: e.g. (A) with $(3, 0, 2)$ gives head $(2, 1, 2) \sim (2, 2, 1)$ and tail $[(3, 1, 1), (4, 1, 0)]$.

4 The Propagation Theorem (G -level)

Theorem 6 (Propagation). *Fix $d \geq 1$. Suppose $G_c(z, q)$ has nonnegative coefficients for every all-positive profile c (i.e. $c_0, c_1, c_2 \geq 1$) with $c_0 + c_1 + c_2 = d$. Then $G_c(z, q)$ has nonnegative coefficients for every profile of weight d .*

Proof. We prove nonnegativity of $[z^n q^N]G_c$ by induction on n , and for fixed n by strong induction on the head-chain. Note first that each F_c , hence each G_c , has constant term $[z^0]G_c = 1 \geq 0$ (setting $z = 0$ kills all bounded-max weights except the empty partition after the $(zq; q)_\infty$ twist; formally $G_c(0, q) = F_c(0, q) = 1$). So take $n \geq 1$.

Let c be zero-containing and apply its R -relation (2):

$$[z^n]G_c(z) = q^n [z^n]G_{\text{head}(c)} + \sum_{i=1}^L q^i q^{(i+1)(n-1)} [z^{n-1}]G_{b_i},$$

where $[z^m]G(zq^s) = q^{sm}[z^m]G(z)$ was used. Every tail term involves $[z^{n-1}]$, nonnegative by induction on n (the b_i may be arbitrary profiles, including c itself: only smaller n is referenced, so self-reference is harmless). For the head term: iterating $c \mapsto \text{head}(c)$ strictly decreases the multiset of parts in a suitable order — concretely, in family (A) the head $(a-1, 1, b)$ is all-positive; in family (B) the head $(a, b-1, 1)$ is all-positive when $b \geq 2$ and equals $(a, 0, 1) \sim$ family (A) rotation when $b = 1$, whose own head is all-positive; and $(d, 0, 0)$ has head $(d-1, 1, 0)$, a family-(B) profile. Hence every head chain reaches an all-positive profile in at most 3 steps, where nonnegativity holds by hypothesis. All terms on the right are therefore nonnegative. $\square$

Remark 7. At $d = 5$ the all-positive core is $\{(3, 1, 1), (2, 2, 1)\}$ up to rotation — precisely the profiles for which Warnaar needed explicit multisum formulas. At $d = 8$ the core is the 7 orbits $(6, 1, 1), (5, 1, 2), (4, 1, 3), (4, 3, 1), (5, 2, 1), (4, 2, 2), (3, 3, 2)$. The propagation mechanism itself carries no $d = 5$ -specific input.

5 Corollary: the case $d = 2$ solved exactly

Corollary 8. $Q_n^{(1,1,0)}(q) = q^{n^2}$ and $Q_n^{(2,0,0)}(q) = q^{n(n+1)}$ for all $n \geq 0$. In particular the positivity conjecture holds at $d = 2$, and $Q_n(1) = 1 = (\frac{1}{6} \cdot 3 \cdot 4 - 1)^n$.

Proof. At $d = 2$ there are no all-positive profiles, and the R -relations close on themselves. Writing $g_c(n) := [z^n]G_c$, (0) gives $g_{(2,0,0)}(n) = q^n g_{(1,1,0)}(n)$, and (B) with $(a, b) = (1, 1)$ gives head $(1, 0, 1) \sim (1, 1, 0)$ (rotation) and tail $[(0, 1, 1)] \sim [(1, 1, 0)]$:

$$g_{(1,1,0)}(n) = q^n g_{(1,0,1)}(n)|_{\text{shift}} + q^{2n-1} q g_{(1,1,0)}(n-1)$$

which upon careful bookkeeping of the z -shifts yields the first-order recursion $g_{(1,1,0)}(n) = \frac{q^{2n-1}}{1-q^n} g_{(1,1,0)}(n-1)$, solved by $g_{(1,1,0)}(n) = q^{n^2}/(q; q)_n$. Multiplying by $(q; q)_n$ gives $Q_n^{(1,1,0)} = q^{n^2}$ and then $Q_n^{(2,0,0)} = q^n \cdot q^{n^2} \cdot \frac{(q; q)_n}{(q; q)_n} = q^{n(n+1)}$. Both closed forms were verified against the raw Corteel–Welsh recursion for $n \leq 6$ at series precision 520. $\square$

6 The bounded level: exact identity and the INJ inequality

Extracting $[z^n]$ from (2) and multiplying by $(q^\ell; q^\ell)_n$ gives the *exact* cross-profile recursion (using $(q^\ell; q^\ell)_n = (1 - q^{\ell n})(q^\ell; q^\ell)_{n-1}$):

$$Q_n^c = q^n Q_n^{\text{head}(c)} + (1 - q^{\ell n}) \sum_{i=1}^L q^{(i+1)n-1} Q_{n-1}^{b_i}. \quad (3)$$

The factor $(1 - q^{\ell n})$ is not positive, so (3) does *not* immediately propagate positivity — this is the precise point where naive Q -level inheritance fails, and computation confirms it genuinely fails for some R -relation variants (e.g. the long-tail variant of $(d-1, 1, 0)$ fails at q^4 for $n = 2$, at every tested d).

Definition 9 (Injection inequality). An R -relation for c satisfies (INJ_c) if, coefficientwise,

$$Q_n^{\text{head}(c)} \geq \sum_{i=1}^L q^{(i+1)n-1} Q_{n-1}^{b_i} \quad \text{for all } n \geq 1.$$

Proposition 10. Write $\Sigma := \sum_{i=1}^L q^{(i+1)n-1} Q_{n-1}^{b_i}$ and $\Sigma' := (1 + q^n + \dots + q^{(\ell-1)n}) \Sigma$. If the R -relation for c satisfies the (ℓ -strengthened) inequality $Q_n^{\text{head}} \geq \Sigma'$ coefficientwise, and $Q_{n-1}^{b_i} \geq 0$ for all i , then $Q_n^c \geq 0$.

Proof. Since $1 - q^{\ell n} = (1 - q^n)(1 + q^n + \dots + q^{(\ell-1)n})$, identity (3) reads $Q_n^c = q^n Q_n^{\text{head}} + (1 - q^n) \Sigma'$. Rearranging,

$$Q_n^c = q^n (Q_n^{\text{head}} - \Sigma') + \Sigma',$$

and both summands are coefficientwise nonnegative by hypothesis ($\Sigma' \geq 0$ because each $Q_{n-1}^{b_i} \geq 0$). For $\ell = 1$ this is exactly (INJ_c) as defined above; for $\ell = 3$ the scan tests the strengthened version. $\square$

Theorem 11 (Conditional Q -level propagation). Fix d . Assume (i) $Q_n^c \geq 0$ for all all-positive (core) profiles c and all n , and (ii) every zero-containing orbit admits an R -relation variant satisfying (INJ_c) . Then $Q_n^c \geq 0$ for all profiles of weight d , by induction on n and the head-chain as in Theorem 6, using Proposition 10 at each zero-containing profile.

Computational status of (ii). For every zero-containing orbit at $d \in \{4, 5, 6, 7, 8, 10, 11\}$ (including $3 \mid d$ with the $(q^3; q^3)_n$ prefactor), an INJ-satisfying variant exists and was verified for $n \leq 4$ (resp. $n \leq 5$ at $d = 5, 8$) at series precision ≥ 420 : a clean sweep with zero failures. The unique delicate orbit is $(d-1, 1, 0)$, whose long-tail variant fails INJ at the d -independent exponents q^4, q^8, q^{14}, q^{21} ($n = 2, 3, 4, 5$) but whose short-tail variant (head $(d-2, 2, 0)$, tail $[(d-2, 1, 1), (d-1, 1, 0)]$) satisfies it.

7 A worked example: $d = 8$, profile $(6, 0, 2)$

The R -relation from Theorem 3(A) is

$$G_{(6,0,2)}(z) = G_{(5,1,2)}(zq) + zq G_{(6,1,1)}(zq^2) + zq^2 G_{(7,1,0)}(zq^3),$$

verified as an exact q -series identity for $[z^n]$, $n \leq 5$, at precision 520 against the raw alternating-sign Corteel–Welsh solution. At the bounded level,

$$Q_n^{(6,0,2)} = q^n Q_n^{(5,1,2)} + (1 - q^n)(q^{2n-1} Q_{n-1}^{(6,1,1)} + q^{3n-1} Q_{n-1}^{(7,1,0)}),$$

and (INJ): $Q_n^{(5,1,2)} \geq q^{2n-1} Q_{n-1}^{(6,1,1)} + q^{3n-1} Q_{n-1}^{(7,1,0)}$ holds coefficientwise for $n \leq 5$.

8 Conclusions and open ends

1. **GREEN.** G -level propagation (Theorems 3, 6): positivity for all profiles reduces to the all-positive core, at every d . The mechanism generalises Warnaar/CDU's $d = 5$ equations verbatim.
2. **GREEN.** $d = 2$ solved exactly (Corollary 8).
3. **YELLOW.** Q -level propagation is conditional on the injection inequality INJ, verified computationally for $d \leq 11$ but unproved. Proving INJ for the short-tail variants (two-term inequalities) is the natural next target.
4. **Open.** The core itself: all-positive profiles $((4, 2, 2), (3, 3, 2), (5, 2, 1), (4, 3, 1), \dots)$ at $d = 8$ still require positive formulas — Warnaar's Conjecture 2 and Kanade–Russell Conjecture 5.1 cover parts of it.